Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges_1.csv

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This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyse historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.
Modules Used:
The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:
In this scenario, railway gauge data is imported from a CSV file into a Pandas Data Frame. Display first few rows and columns. Check and replace Missing values with 0 to ensure data consistency. Finally, relevant columns are converted into numeric data types to enable accurate analysis and visualization in later stages.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

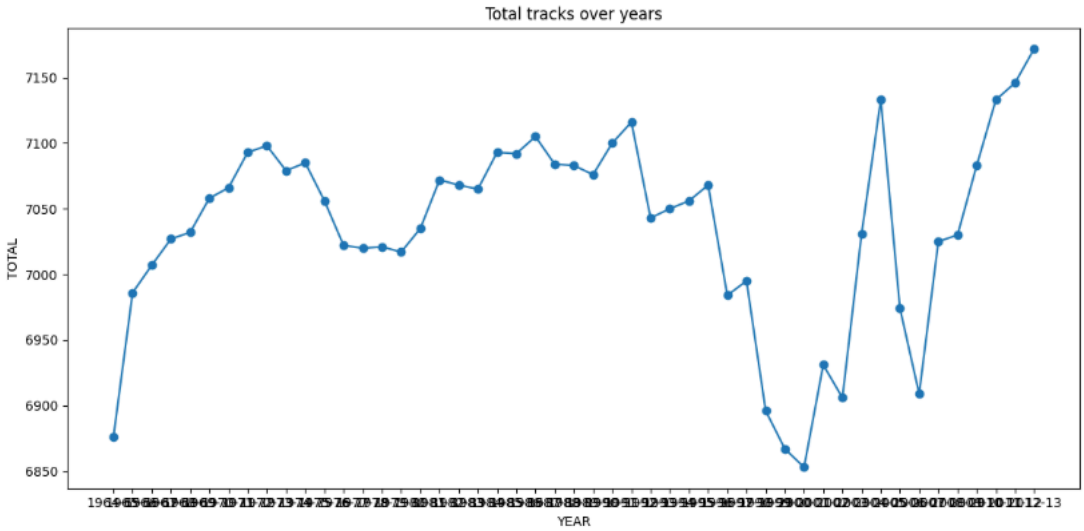
Conclusion:
The dataset was successfully loaded and prepared for analysis by handling missing values and ensuring proper data types. This cleaning process improves data consistency and reliability. Overall, the dataset is now structured and ready for deeper analysis in the subsequent scenarios.

Scenario 2: Simple Visualization (🟢 Easy)

Description:
This scenario focuses on visualizing the growth of the railway network over time using a simple line graph. And Year and Total columns are extracted from the dataset for analysis.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph:
The graph is a **line plot** with Year on the X-axis and Total railway track length on the Y-axis. Each data point is marked and connected to show changes over time. The chart clearly illustrates the progression of the railway network, making it easy to observe increases or decreases across years.



Conclusion:

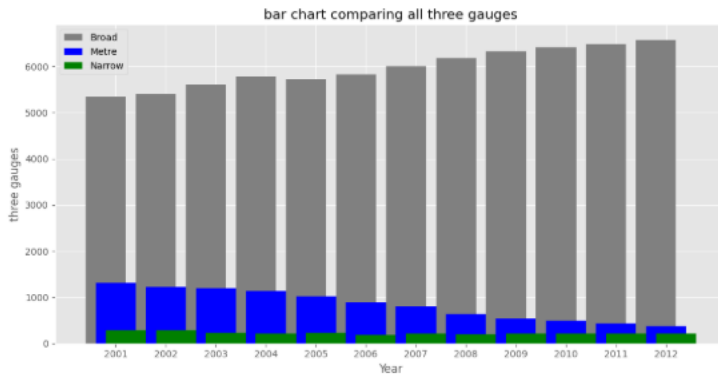
The line graph shows that the total railway network generally follows an increasing trend over the years, indicating continuous expansion. Although there may be minor fluctuations or periods of slower growth, the overall direction remains upward.
The visualization provides a clear understanding of long-term growth patterns in the railway system.

Scenario 3: Filtering + Bar Chart (🟡 Medium)

Description:
Filter the dataset to get only years after 2000. And select gauges columns from filtered data. After that, plot grouped bar chart which compares three gauges for selected years. Finally, able to Identify the dominating gauge after 2000.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph:
The graph is a **grouped bar chart** where each year is represented on the X-axis, and the track length is shown on the Y-axis. For every year, three bars are displayed side by side representing Broad Gauge, Metre Gauge, and Narrow Gauge. The chart makes it easy to compare the contribution of each gauge over time.



Conclusion:
The analysis clearly shows that Broad Gauge dominates the railway network in years after 2000. Its values are significantly higher compared to Metre and Narrow Gauges. Meanwhile, Metre and Narrow Gauges show a declining or minimal presence in recent years. This indicates a shift toward standardization, with increasing preference for Broad Gauge. Overall, the data reflects modernization and consolidation of the railway system.

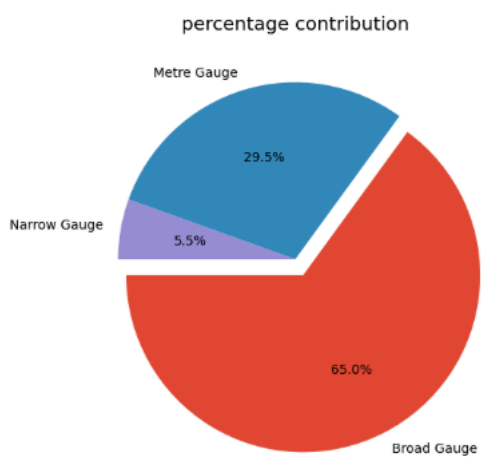
Scenario 4: Feature Engineering + Pie Chart (🟡 Medium)

Description:
This scenario focuses on feature engineering to understand the overall contribution of different railway gauges. The total sum of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated across all years in the dataset. These aggregated values are stored in a new structure for easy analysis. A pie chart is then used to represent the proportion of each gauge type. This visualization helps in understanding the historical dominance of each gauge in the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').

5.	nterpret which gauge contributes the most to the entire history of the railway network.

Graph:
The graph is a **pie chart** displaying the percentage contribution of Broad Gauge, Metre Gauge, and Narrow Gauge. Each slice represents one gauge type, with percentage labels shown using `autopct`. The BroadGauge slice is slightly highlighted (exploded) to emphasize its contribution. The chart provides a clear visual comparison of how much each gauge contributes to the total railway network.



Conclusion:
The pie chart reveals that Broad Gauge contributes the largest share to the overall railway network. Its percentage is significantly higher compared to Metre and Narrow Gauges. Metre Gauge holds a smaller portion, while Narrow Gauge contributes the least. This indicates that, historically, Broad Gauge has been the most important component of the railway system. The analysis highlights the dominance and long-term preference for Broad Gauge infrastructure.

Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

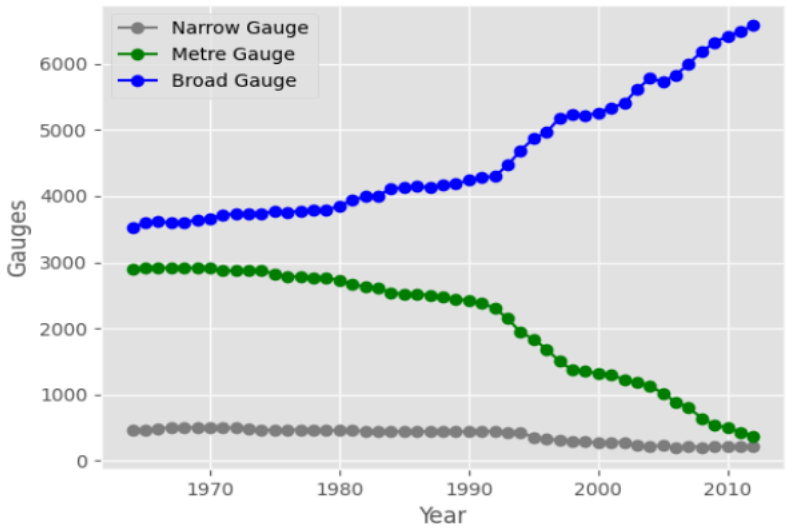
Description:
This scenario focuses on advanced analysis of railway data by combining feature engineering, numerical computation, and multiple visualizations. New percentage-based columns are created to understand the composition of each gauge relative to the total network. NumPy is used to calculate yearly growth, helping identify trends and fluctuations. Multiple graphs, including line plots and stacked bar charts, are used to present both proportional and absolute

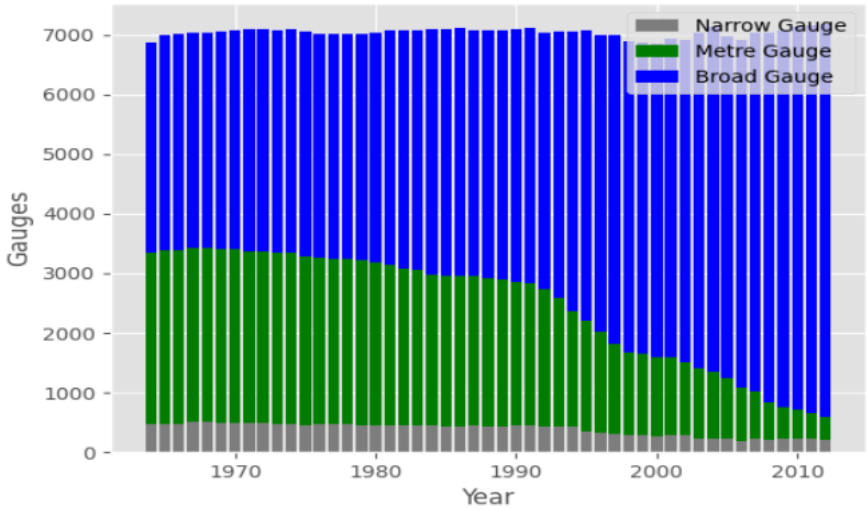
data. This approach provides a deeper understanding of long-term railway development patterns.

Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graph:

Two types of graphs are used in this analysis. The first is a **line graph** showing the percentage composition of Broad, Metre, and Narrow Gauges over time, which helps visualize how each gauge's share changes. The second is a **stacked bar chart** representing the total composition of all three gauges for each year, showing how they collectively form the entire railway network. Together, these graphs provide both relative and absolute perspectives of the data.





Conclusion:

The analysis shows that Broad Gauge has consistently increased both in absolute terms and percentage composition over time. The percentage line graph clearly indicates a steady rise in Broad Gauge, while Metre and Narrow Gauges decline significantly. The yearly growth analysis highlights specific years with strong expansion, along with occasional periods of slower growth or decline. Overall, the data strongly suggests a transition toward a standardized railway system. Broad Gauge has become the dominant and preferred choice for expansion and modernization.

Final Conclusion:

Across all scenarios, the railway dataset reveals a clear pattern of growth, transformation, and standardization. Initial data cleaning ensured accurate analysis, while visualizations helped uncover trends in total track expansion. Comparative analysis showed that Broad Gauge increasingly dominates over time, replacing Metre and Narrow Gauges. Advanced analysis confirmed that the railway system is steadily shifting toward a single, unified gauge. This reflects strategic planning aimed at improving efficiency, capacity, and connectivity in the railway network.

What you have learned

Through this project, I learned how to perform complete data analysis using Python libraries like NumPy, Pandas, and Matplotlib. I understood how to clean and preprocess raw data, handle missing values, and convert data types. I gained experience in creating different types of visualizations such as line graphs, bar charts, pie charts, and stacked charts. I also learned how to extract meaningful insights, identify trends, and interpret results effectively. Overall, this project helped me understand how data analysis can be used to support real-world decision-making.